

Shiv Nadar University Chennai
CS3807 Deep Learning Laboratory
Experiment 4
Comparative Study of Deep Convolutional Neural
Network Architectures Using Transfer Learning

Degree & Branch — B.Tech Artificial Intelligence & Data Science
— Semester V
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1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogLeNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogLeNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogLeNet (Inception Network)

GoogLeNet, proposed by Szegedy et al. in 2014, introduced the Inception Module, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogLeNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the vanishing gradient problem, making optimization difficult. ResNet solves this issue using skip (residual) connections, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$\text{Output} = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

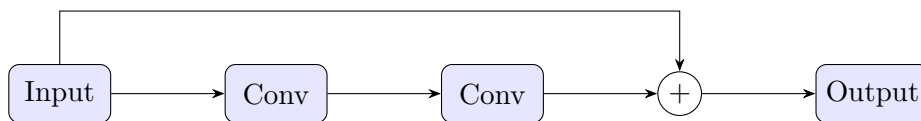


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.

4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

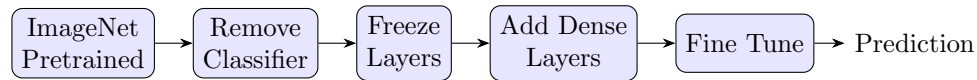


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images: 50,000
- Testing Images: 10,000
- Number of Classes: 10
- Image Size: $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0,1]$.
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet 50
- MobileNet V2
- EfficientNet B0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10-20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5-10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

Source code:

GitHub Repository: https://github.com/un-shreeya/DeepLearning_Lab

17. Mandatory Plots & Inferences



Figure 3: Sample images from the CIFAR-10 dataset.

Inference: The sample batch shows images across 10 distinct classes. Since CIFAR-10 images are quite small (32x32 pixels), fine features like eyes or small patterns are blurry, making it important to upsample the inputs when passing them into deep backbones like ResNet.

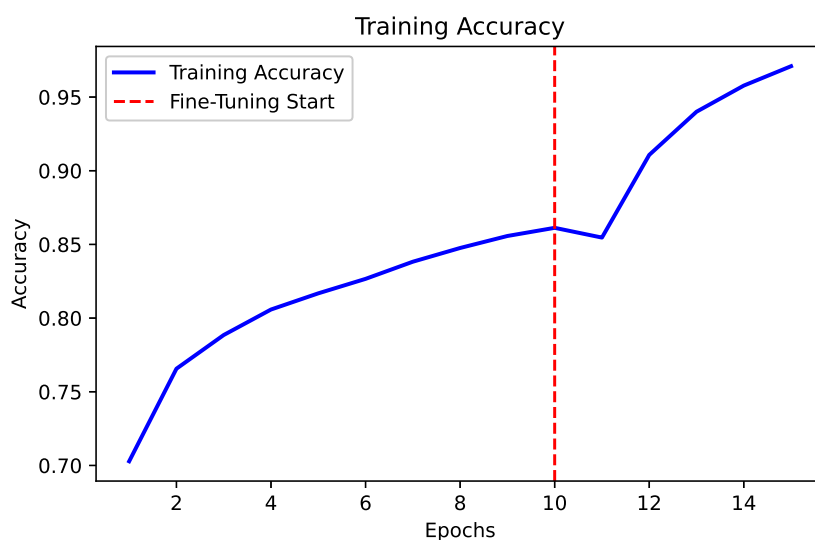


Figure 4: Training Accuracy vs. Epochs.

Inference: Training accuracy increases during first 10 epochs while the feature extractor is frozen. Once finetuning starts at epoch 10, accuracy increases from around 86% up to 97% by epoch 15.

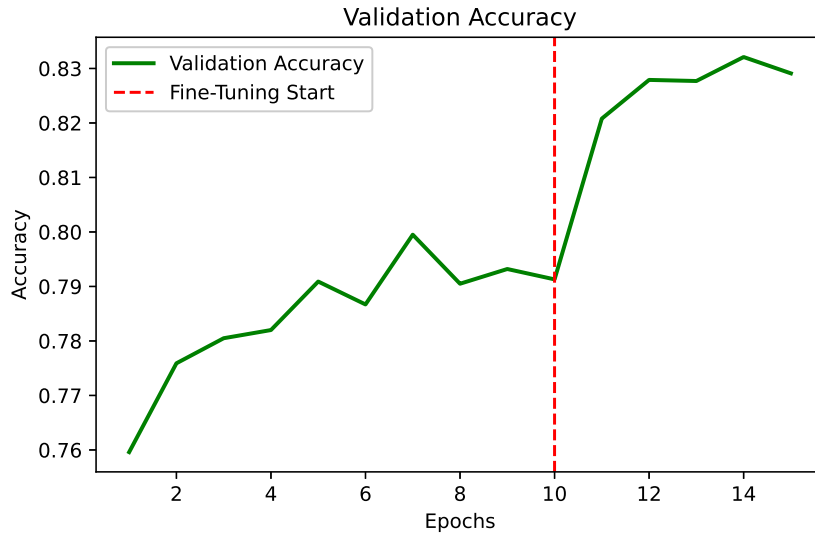


Figure 5: Validation Accuracy vs. Epochs.

Inference: The validation accuracy for the first feature extraction levels off at about 79%. When the top layers are unfrozen, there's a increase. Thus validation accuracy increases to its maximum of around 83-84%.

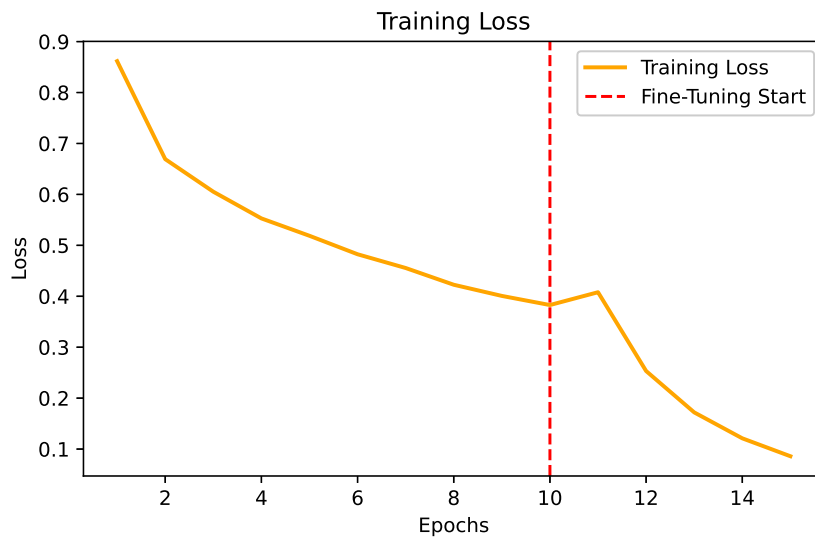


Figure 6: Training Loss vs. Epochs.

Inference: Training loss drops from around 0.88 to around 0.38 during initial frozen phase. There's a small increase at epoch 11 right when fine-tuning begins, but it drops right after, to below 0.10 (the network starts adapting).



Figure 7: Validation Loss vs. Epochs.

Inference: Validation loss fluctuates between 0.60 and 0.67 in the 1st phase, drops to 0.56 right after unfreezing the conv. layers, but begins rising back up to 0.68 by epoch 15. Keeping finetuning open past 3–4 epochs results in the training set starting to overfit.

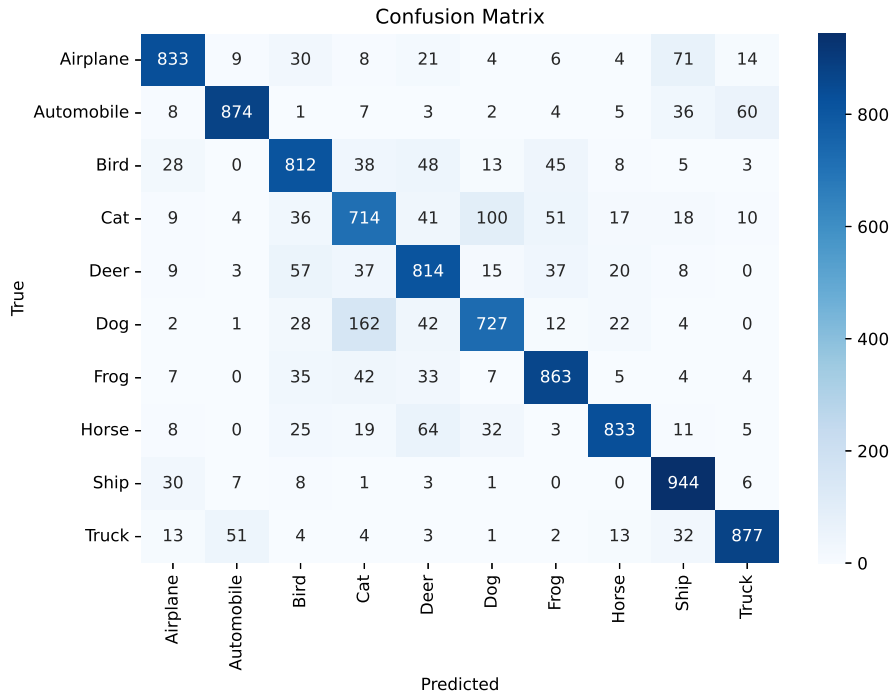


Figure 8: Confusion Matrix of ResNet50 on the CIFAR-10 Test Set.

Inference: The model performs really well on vehicles like Ships (944 correct), Trucks (877), and Automobiles (874) due to their clear straight lines and borders. Errors mostly happen between visually similar animals- for eg. Dogs are mistaken for Cats (162 times) and Cats are mistaken for Dogs (100 times).



Figure 9: Sample Misclassified Test Images.

Inference: Misclassification of images usually happen when the images are partially cropped, viewed at unusual angles, or blending into a busy background (such as animals in grass). At 32x32 resolution, these background textures easily override low-level feature cues.

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	97.09%
Testing Accuracy	82.91%
Precision	0.8313
Recall	0.8291
F1-score	0.8292
Training Time	1177.13 seconds
Total Parameters	23,851,274

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	121,182	58.79%	19.91s
AlexNet	4,026,954	67.68%	56.86s
VGG16	14,781,642	77.61%	195.61s
GoogleNet	22,066,346	69.91%	85.54s
ResNet50	23,851,274	78.41%	109.62s

19. Discussion Questions

1. Why is AlexNet considered a breakthrough in deep learning?

It proved that deep NNs could outperform traditional computer vision models. It uses ReLU to speed up training compared to Sigmoid/Tanh, applies Dropout to prevent overfitting, and implements GPUs for parallel computation.

2. Why does VGG16 use only 3×3 convolution filters?

Stacking 2 3x3 filters covers the same effective receptive field as a single 5x5 filter, and 3 3x3

filters covers the same effective receptive field as that of a 7x7 filter. this way, the trainable parameters used can be significantly reduced, while also adding extra ReLU non-linearities.

3. Explain the advantages of the Inception module.

It executes 1x1, 3x3, 5x5 convolutions and max pooling simultaneously to obtain multi-scale feature representation. It also uses 1x1 convolutions as bottleneck layers before larger convolutions to reduce channel depth (saves computational cost).

4. What is the purpose of residual learning?

It solves the vanishing gradient problem in extremely deep networks using residual skip connections. Instead of directly learning a target mapping $H(x)$, it learns the residual mapping $F(x) = H(x) - x$, allowing gradients to flow backward through the identity shortcut without diminishing.

5. Differentiate LeNet and ResNet.

LeNet-5 is a shallow, 7-layer network (2 convol. layers, 2 avg. pooling layers, 3 fully connected layers) designed for simple 32x32 grayscale digit recognition - uses Sigmoid/Tanh activations). ResNet is a modern, deep architecture (50 to 152+ layers) for complex RGB classification - uses residual skip connections, batch normalization, and ReLU activations.

6. What is Transfer Learning?

It is a technique where a pretrained model (trained on a massive dataset like ImageNet) is chosen instead of training a new model from scratch. Since its pretrained, base layers already have basic knowledge about universal features like edges and textures. It saves significant training time and reduces data requirements.

7. Why is fine tuning required?

Feature extraction keeps the pretrained convolutional base layers frozen and only trains the new classifier head. Finetuning unfreezes the top convolutional layers and trains them at very low learning rate to adapt high level features of the target domain.

8. Explain the difference between dilated convolution and transpose convolution.

Dilated convolution expands the receptive field by inserting gaps between (kernel) weights without increasing number of trainable parameters (makes it ideal for dense context aggregation). Transpose convolution performs learnable upsampling - it expands feature map spatial dimensions.

9. Why do pretrained models converge faster?

A model trained from zero has random weights. It has to spend the first few epochs figuring out basic visual concepts like edges, curves, color gradients, and textures. A pretrained model (like one trained on ImageNet) already has basic knowledge about optimal weights, edges, outlines, etc.. Since the feature extraction is already configured, the optimizer only needs to tweak the final layers to recognize the specific classes.

10. Compare the computational complexity of LeNet and ResNet.

LeNet-5 has 121K parameters. Shallow 7-layer architecture (2 convolutional layers + 2 average pooling layers + 3 fully connected layers). It processes operations sequentially without shortcut paths and can easily run on standard CPUs. ResNet50 has 25.6M parameters. It has a deep 50 layer architecture. it uses residual skip connections and bottleneck design (1×1 , 3×3 , 1×1 convolutions). It uses GPU hardware acceleration for parallel processing, and batch normalization to train efficiently.

20. Additional Exercises

1. Implement transfer learning using VGG16.

Load the pretrained model, add Global Average Pooling layer, a dense ReLU activation layer, and Softmax output layer. During the initial feature extraction phase, base weights should be frozen.

2. Repeat the experiment using ResNet50.

ResNet inputs are upsampled to 96x96. Train with frozen base weights for 10 epochs, then unfreeze the top residual blocks for 5 extra (finetuning) epochs to boost accuracy.

3. **Compare Adam and SGD optimizers.**

Adam converges faster because it automatically adjusts learning rates for each parameter using adaptive momentum. SGD with momentum takes longer to converge as it needs careful learning rate tuning. But SGD can generalize slightly better by settling into flatter minima.

4. **Compare training with frozen layers and fine tuning.**

Training only with frozen layers is faster and stops the model from overfitting on small datasets, but model doesn't reach its max accuracy. Finetuning unfreezes top convolutional layers with a lower learning rate, letting high-level features adapt to the target dataset to improve performance.

5. **Evaluate the model using Precision, Recall and F1-score.**

Calculate macro and weighted averages using `sklearn.metrics.classification_report`. These metrics highlight class specific performance, revealing if visually similar categories (like Cat vs. Dog) suffer from high false positive or false negative rates.

6. **Compare the performance of LeNet, AlexNet and ResNet.**

LeNet-5 scored 58.79% accuracy (due to its small 7-layer architecture). AlexNet scored 67.68% accuracy (by introducing ReLU and Dropout). ResNet50 scored the highest accuracy, 78.41% during frozen feature extraction. After finetuning, it scored 82.91%.

21. References

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